Ulster University

COM 682-Week 11

Cloud Native Development

- # Advanced Azure Services –Cognition

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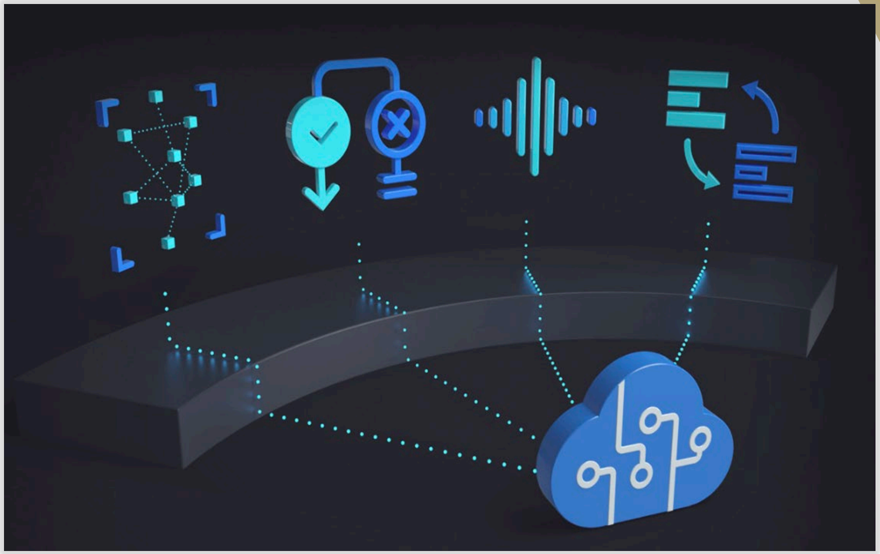
Overview

What is Azure Cognitive Services?

- Content Safety
  - Prompt Shields
  - Text Analyser
  - Image Analyser
- Language
  - Named Entity Recognition (NER)
  - Custom Question Answering (Q&A Maker)
  - Text Analytics
  - Text analytics for health
- Speech
  - Speech to Text
  - Text to Speech
  - Speech Translation
  - Speaker Recognition
- Vision
  - Computer Vision
  - Custom Vision
  - Face API



Content Safety

Azure Cognitive Services

Azure Cognitive Services

Content Safety

Azure AI Content Safety is an AI service that detects harmful user-generated and AI-generated content in applications and services.

Azure AI Content Safety includes text and image APIs that allow you to detect material that is harmful.

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Content Safety- Where it's used

The following are a few scenarios in which content moderation service is required:

- User prompts submitted to a generative AI service.
- Content produced by generative AI models.
- Online marketplaces that moderate product catalogs and other user-generated content.
- Gaming companies that moderate user-generated game artifacts and chat rooms.
- Social messaging platforms that moderate images and text added by their users.
- Enterprise media companies that implement centralized moderation for their content.
- K-12 education solution providers filtering out content that is inappropriate for students and educators.

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Content Safety- Prompt Shields

- Prompt Shields in Azure AI Content Safety are designed to safeguard generative AI systems from generating **harmful** or **inappropriate content**.
- These shields detect and mitigate risks associated with both User Prompt Attacks (malicious or harmful user-generated inputs) and Document Attacks (inputs containing harmful content embedded within documents).
- The primary objectives of the "Prompt Shields" feature for GenAI applications are:
  - To detect and block harmful or policy-violating user prompts that could lead to unsafe AI outputs.
  - To identify and mitigate document attacks where harmful content is embedded within user-provided documents.
  - To maintain the integrity, safety, and compliance of AI-generated content, preventing misuse of GenAI systems.

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Content Safety- Text Analyser

- **Moderate text content**
  - Used for evaluating different content moderation scenarios.
  - It takes into account various factors such as the type of content, the platform's policies, and the potential impact on users.

Safe content

Chopping tomatoes and cutting them into cubes or wedges are great ways to practice your knife skills.

Violent content with misspelling

The dog was given a eutanasa injection due to their severed leg bleding profusely from deep lacarations to the lower extremities,...

Multiple risk categories in one sentence

A 51-year-old man was found dead in his car. There were blood stains on the dashboard and windscreen. At autopsy, a...

Multiple languages in one sentence

Painfully twist his arm then punch him in the face jusqu'à ce qu'il perde connaissance.

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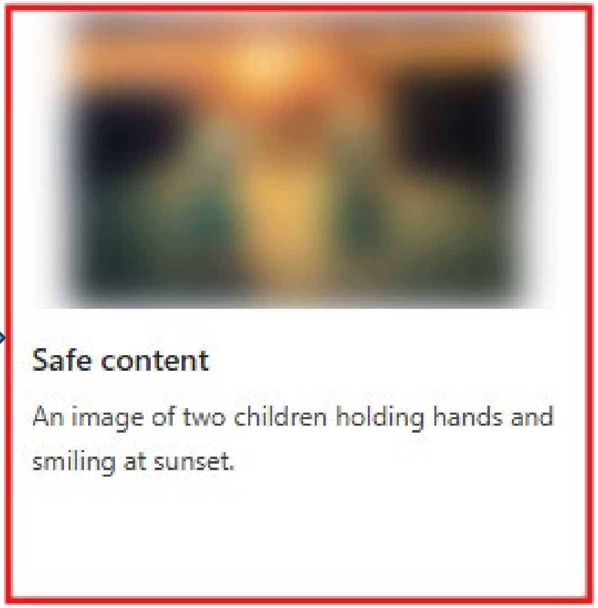
Content Safety- Image Analyser

- **Moderate Image content**
  - Used for evaluating different content moderation scenarios.
  - It takes into account various factors such as the type of content, the platform's policies, and the potential impact on users.



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Content Safety- Image Analyser

- **Moderate Image content**
  - Used for evaluating different content moderation scenarios.
  - It takes into account various factors such as the type of content, the platform's policies, and the potential impact on users.

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Content Safety- Image Analyser

- **Moderate Image content**
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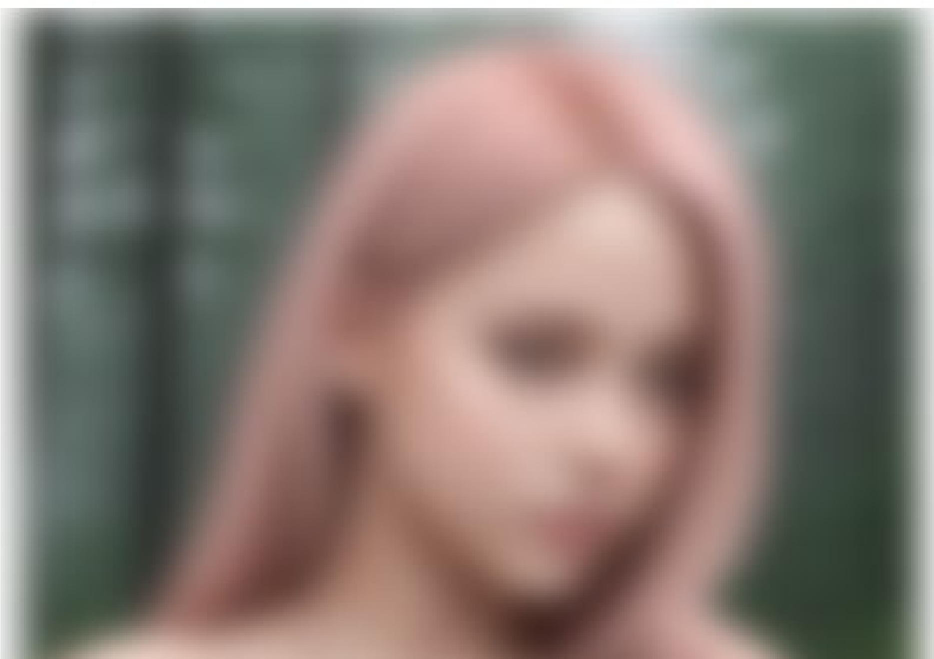

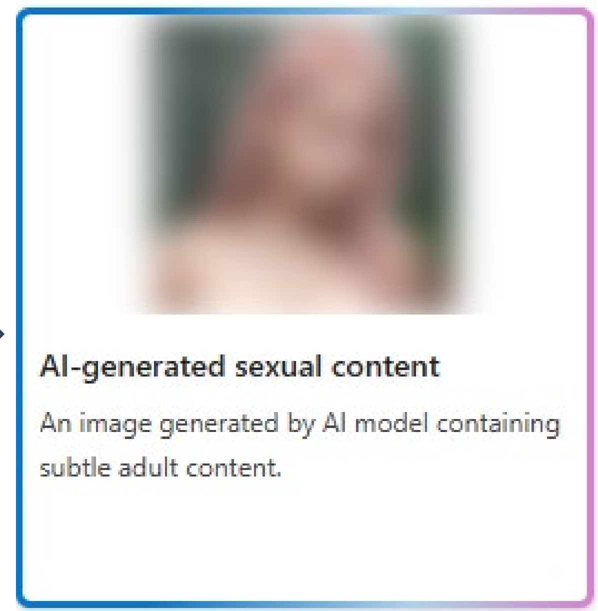

Language

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Azure AI Language

- Azure AI Language is a cloud-based service that provides Natural Language Processing (NLP) features for understanding and analyzing text.
- Use this service to help build intelligent applications using the web-based Language Studio, REST APIs, and client libraries.

➢ **Features:**

- The Language service provides several new features, which can either be:
  - Preconfigured, which means the AI models that the feature uses are not customizable. You just send your data and use the feature's output in your applications.
  - Customizable, which means you'll train an AI model using our tools to fit your data specifically.

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Language *–Named Entity Recognition (NER)*

- Named Entity Recognition (NER) is one of the features offered by Azure AI Language.
- It's a collection of machine learning and AI algorithms in the cloud for developing intelligent applications that involve written language.
- The NER feature can identify and categorize entities in unstructured text. For example: people, places, organizations, and quantities.

Event

Entity value: trip
Confidence: 74.00%

Location

GPE

Entity value: Seattle
Confidence: 100.00%

DateTime

DateRange

Entity value: last week
Confidence: 80.00%

Original text

I had a wonderful trip to Seattle last week.

Event Location DateTime

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Language *–Named Entity Recognition (NER)*

- Scenarios
  - **Enhance search capabilities and search indexing** - Customers can build knowledge graphs based on entities detected in documents to enhance document search as tags.
  - **Automate business processes** - For example, when reviewing insurance claims, recognized entities like name and location could be highlighted to facilitate the review. Or a support ticket could be generated with a customer's name and company automatically from an email.
  - **Customer analysis** – Determine the most popular information conveyed by customers in reviews, emails, and calls to determine the most relevant topics that get brought up and determine trends over time.

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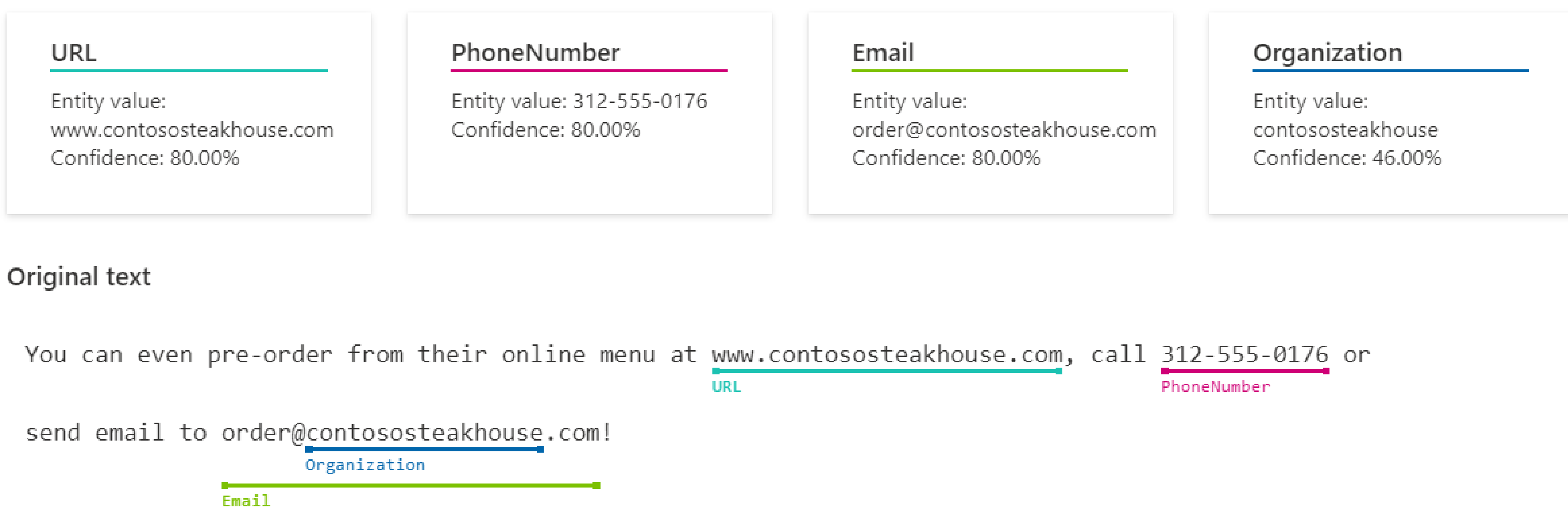
Language –*Personally identifying (PII) and health (PHI) information detection*

- PII detection is one of the features offered by Azure AI Language.
- It's a collection of machine learning and AI algorithms in the cloud for developing intelligent applications that involve written language.
- **The PII detection** feature can identify, categorize, and **redact** (editing or revising) sensitive information in unstructured text. For example: phone numbers, email addresses, and forms of identification.



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Language *–Personally identifying (PII) and health (PHI) information detection*

- **Scenarios**
  - **Apply sensitivity labels -** For example, based on the results from the PII service, a public sensitivity label might be applied to documents where no PII entities are detected. For documents where US addresses and phone numbers are recognized, a confidential label might be applied. A highly confidential label might be used for documents where bank routing numbers are recognized.
  - **Redact some categories of personal information from documents that get wider circulation -** For example, if customer contact records are accessible to frontline support representatives, the company can redact the customer's personal information besides their name from the version of the customer history to preserve the customer's privacy.

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Language *–Personally identifying (PII) and health (PHI) information detection*

- **Scenarios**
  - **Redact personal information to reduce unconscious bias -** For example, during a company's resume review process, they can block name, address and phone number to help reduce unconscious gender or other biases.
  - **Remove personal information from call center transcription –** For example, if you want to remove names or other PII data that happen between the agent and the customer in a call center scenario. You could use the service to identify and remove them.
  - **Data cleaning for data science -** PII can be used to make the data ready for data scientists and engineers to be able to use these data to train their machine learning models. Redacting the data to make sure that customer data isn't exposed.

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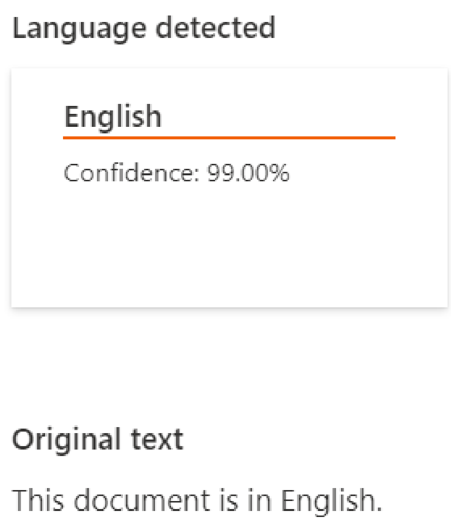
Language –*Language detection*

- **Language detection** is a preconfigured feature that can detect the language a document is written in and returns a language code for a wide range of languages, variants, dialects, and some regional/cultural languages.
- Language detection is able to detect more than **100 languages** in their primary script.
- In addition, it offers **script detection** to detect supported scripts for each detected language according to the **ISO 15924 standard.**



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Language –*Language detection*

- **Language detection features**
  - **Language detection:** Returns one predominant language for each document you submit, along with its ISO 639-1 name, a human-readable name, confidence score, script name and script code according to ISO 15924 standard.
  - **Script detection:** To distinguish between multiple scripts used to write certain languages, such as Kazakh, language detection returns a script name and script code according to the ISO 15924 standard.
  - **Ambiguous content handling:** To help disambiguate language based on the input, you can specify an ISO 3166-1 alpha-2 country/region code. For example, the word "communication" is common to both English and French. Specifying the origin of the text as France can help the language detection model determine the correct language.

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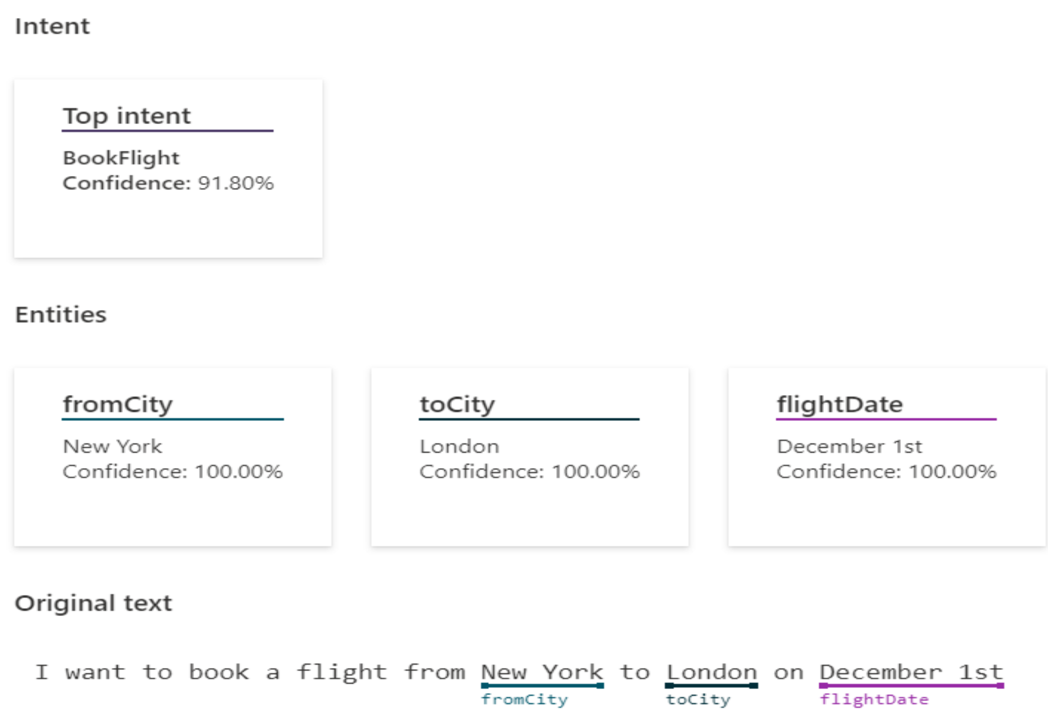
Language –*Conversational language understanding (CLU)*

- **CLU** enables users to build custom natural language understanding models to predict the overall intention of an incoming utterance and extract important information from it.
- CLU only provides the intelligence to understand the input text for the client application and doesn't perform any actions.



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Language –*Conversational language understanding (CLU)*

- **Example usage scenarios**
  - **End-to-end conversational bot:** Use CLU to build and train a custom natural language understanding model based on a specific domain and the expected users' utterances. Integrate it with any end-to-end conversational bot so that it can process and analyze incoming text in real time to identify the intention of the text and extract important information from it.
  - **Human assistant bots:** One example of a human assistant bot is to help staff improve customer engagements by triaging customer queries and assigning them to the appropriate support engineer. Another example would be a human resources bot in an enterprise that allows employees to communicate in natural language and receive guidance based on the query.

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Language *–Conversational language understanding (CLU)*

- **Example usage scenarios**
  - **Command and control application:** When you integrate a client application with a speech to text component, users can speak a command in natural language for CLU to process, identify intent, and extract information from the text for the client application to perform an action. This use case has many applications, such as to stop, play, forward, and rewind a song or turn lights on or off.
  - **Enterprise chat bot:** In a large corporation, an enterprise chat bot may handle a variety of employee affairs. It might handle frequently asked questions served by a custom question answering knowledge base, a calendar specific skill served by conversational language understanding, and an interview feedback skill served by LUIS.

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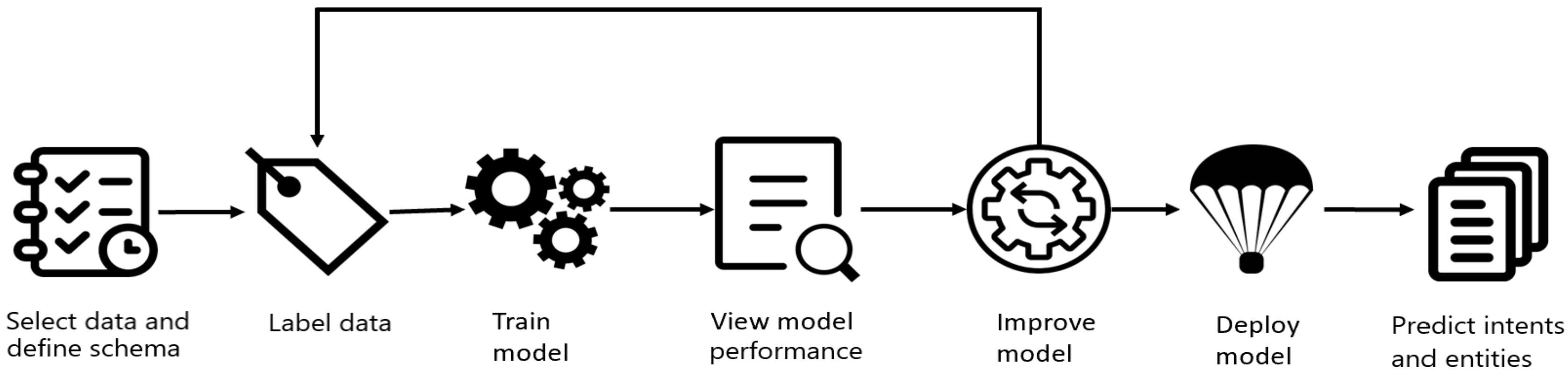
CLU –*Project development lifecycle*



- **Creating a CLU project typically involves several different steps.**
  - **Define your schema:** Know your data and define the actions and relevant information that needs to be recognized from user's input utterances. In this step you create the intents that you want to assign to user's utterances, and the relevant entities you want extracted
  - **Label your data:** The quality of data labeling is a key factor in determining model performance.

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CLU *–Project development lifecycle*

- **Creating a CLU project typically involves several different steps.**
  - **Train the model:** Your model starts learning from your labeled data.
  - **View the model's performance:** View the evaluation details for your model to determine how well it performs when introduced to new data.
  - **Improve the model:** After reviewing the model's performance, you can then learn how you can improve the model.
  - **Deploy the model:** Deploying a model makes it available for use via the Runtime API.
  - **Predict intents and entities:** Use your custom model to predict intents and entities from user's utterances.

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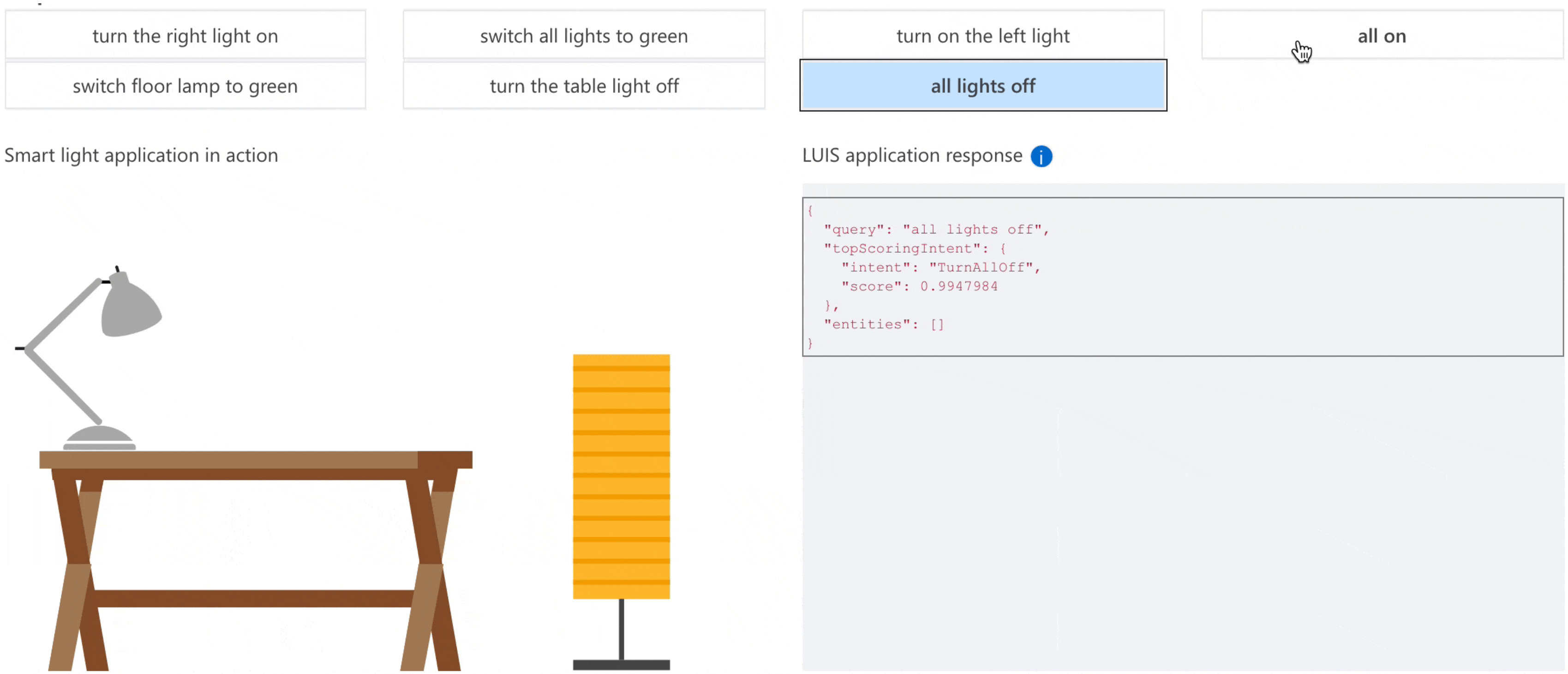
Language –*Language Understanding*

- **Controlling IoT devices using a voice assistant**

Create seamless conversational interfaces that understand natural language with all your internet-accessible devices—from your connected television or fridge to devices in a connected power plant.



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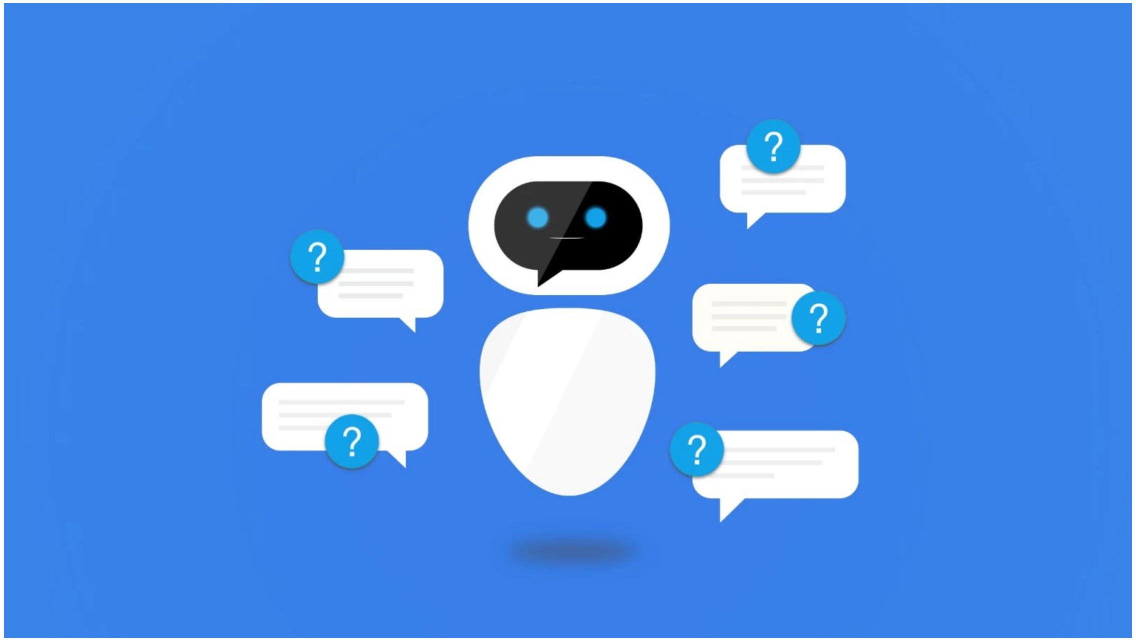
Language *–Custom question answering*

- **Custom question answering** is a cloud-based API service that **lets you create a conversational question-and-answer** layer over your existing data.
- It is used to find appropriate answers from customer input or from a project.
- **Custom question answering** is commonly used to build conversational client applications, which include social media applications, chat bots, and speech-enabled desktop applications.



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Language –*Custom question answering:* ***Capabilities***

- **Custom question answering:** Using this capability users can customize different aspects like edit question and answer pairs extracted from the content source, define synonyms and metadata, accept question suggestions etc.

- **QnA Maker:** This capability allows users to get a response by querying a text passage without having the need to manage knowledge bases.

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Language –*Custom question answering (CQA)*

When to use Custom question answering?

- **When you have static information** - Use CQA Maker when you have static information in your knowledge base of answers.
- **When you want to provide the same answer to a request, question, or command** - when different users submit the same question, the same answer is returned.
- **When you want to filter static information based on meta-information** - add metadata tags to provide additional filtering options relevant to your client application's users and the information. Common metadata information includes chit-chat, content type or format, content purpose, and content freshness.
- **When you want to manage a bot conversation that includes static information** - your knowledge base takes a user's conversational text or command and answers it.

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Language –*CQA*

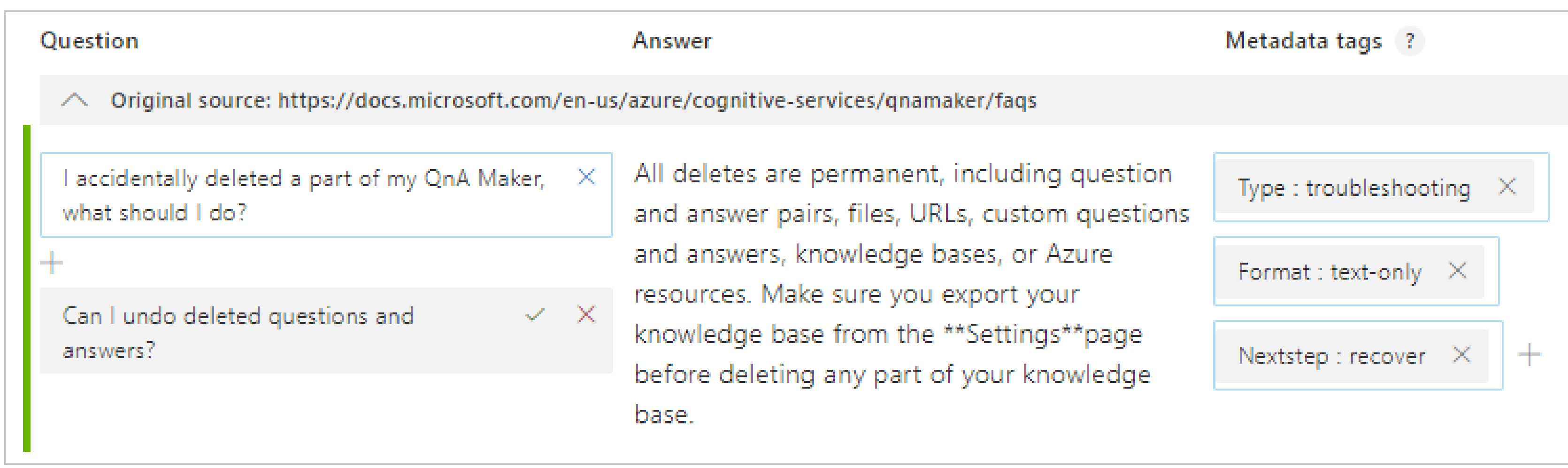
What is a project?

Custom question answering imports your content into a project full of question-and-answer pairs. The import process extracts information about the relationship between the parts of your structured and semi-structured content to imply relationships between the question-and-answer pairs. You can edit these question-and-answer pairs or add new pairs.

After you publish your project, a client application sends a user's question to your endpoint. Your CQA service processes the question and responds with the best answer.

The content of the question-and-answer pair includes:

- All the alternate forms of the question
- Metadata tags used to filter answer choices during the search
- Follow-up prompts to continue the search refinement



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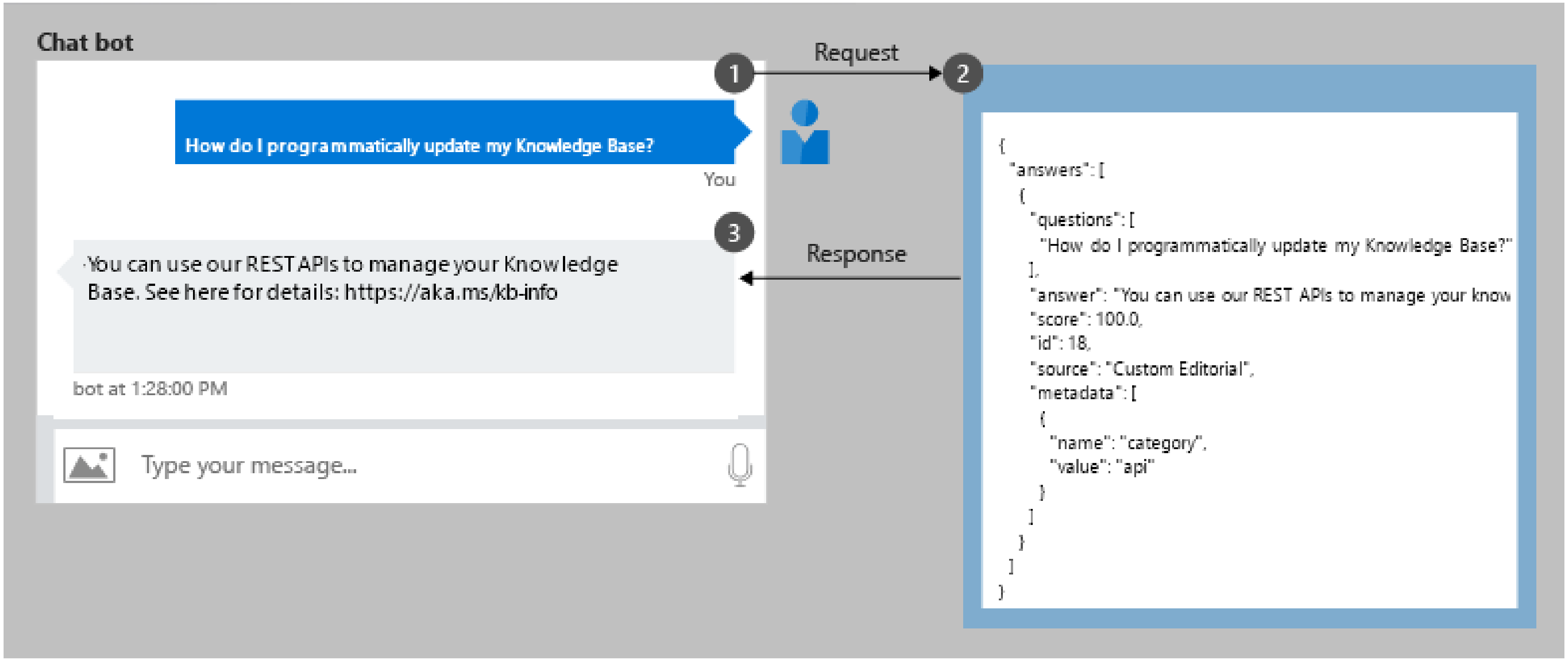
Language –*CQA*

Create a chat bot programmatically

- Once a CQA project is published, a client application sends a question to your project endpoint and receives the results as a JSON response. A common client application for QnA Maker is a chat bot.



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Language –*Text Analytics*

- The Text Analytics API is a cloud-based service that provides **Natural Language Processing (NLP)** features for text mining and text analysis, including:
  - Sentiment analysis
  - Key phrase extraction
  - Text summarization

Azure Cognitive Services

Language –*Text Analytics*

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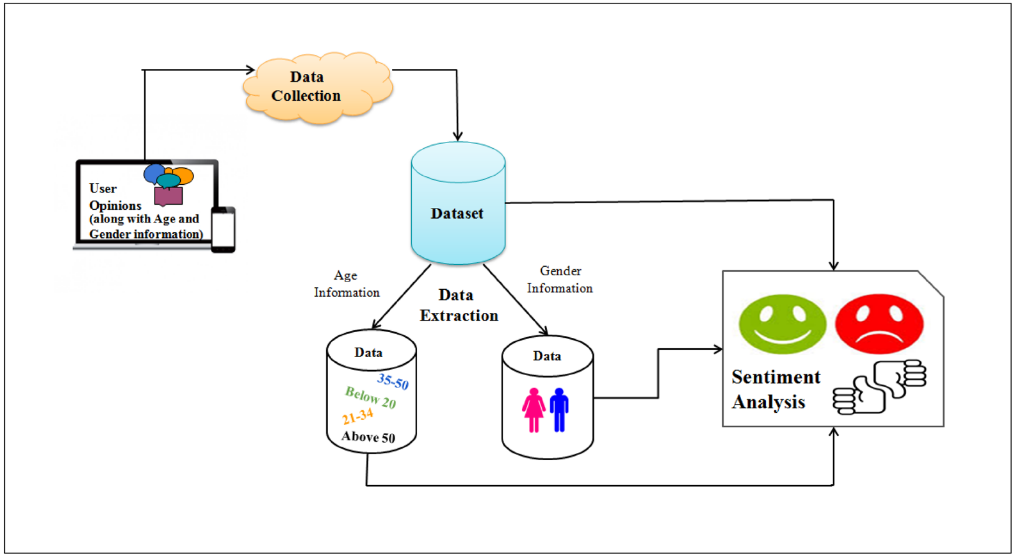

Use sentiment analysis (SA) to process text and classify positive or negative sentiment.

The feature provides sentiment labels (such as "negative", "neutral" and "positive") based on the highest confidence score found by the service at a sentence and document level.

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Language –*Sentiment analysis*

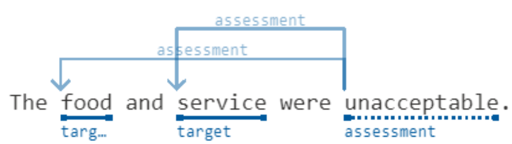
Analyzed sentiment

Document sentiment

Negative
Confidence: 0.00%

0.00%	0.00%	99.00%
POSITIVE	NEUTRAL	NEGATIVE

Sentence 1



Sentence sentiment

Negative
Confidence: 0.00%

0.00%	0.00%	99.00%
POSITIVE	NEUTRAL	NEGATIVE

Opinion

Target: food

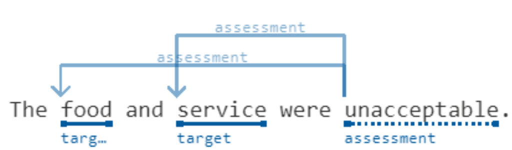
Assessments:
unacceptable (negative, 100.00%)

Opinion

Target: service

Assessments:
unacceptable (negative, 100.00%)

Original text



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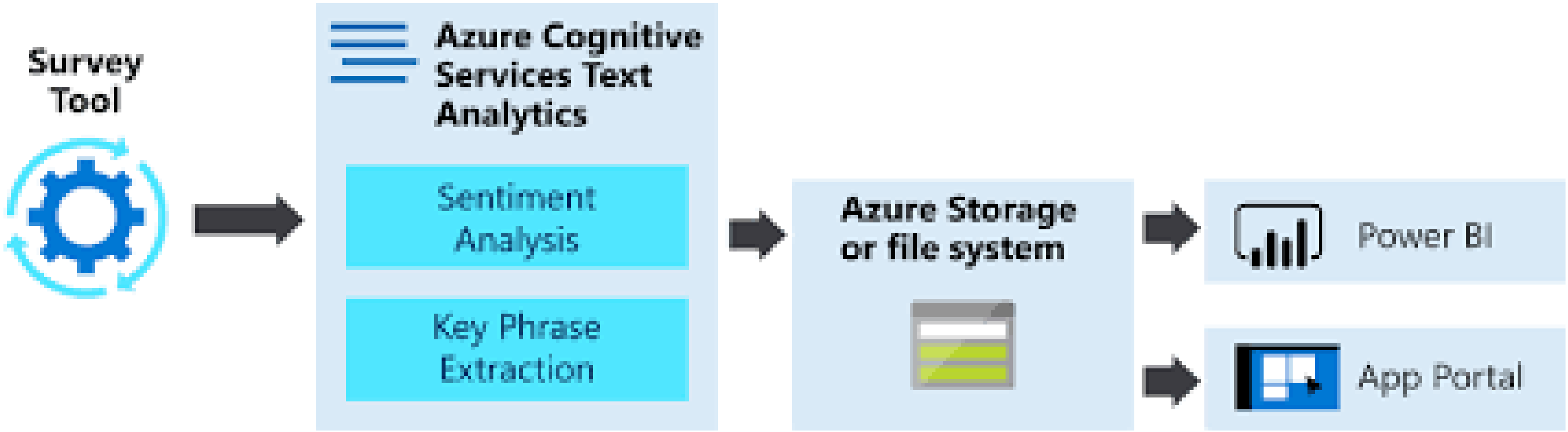
Language –*Text Analytics*

- The Text Analytics API is a cloud-based service that provides **Natural Language Processing (NLP)** features for text mining and text analysis, including:
  - Sentiment analysis
  - **Key phrase extraction**
  - Text summarization

Use key phrase extraction (KPE) to quickly identify the main concepts in text.

For example, in the text **"The food was delicious and there were wonderful staff"**,

Key Phrase Extraction will return the main talking points: **"food"** and **"wonderful staff"**.



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Language –*Key phrase extraction*

Key phrases

Pike place market, favorite Seattle attraction

Original text

Pike place market is my favorite Seattle attraction.

Key phrase   Key phrase

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Language –*Text Analytics*

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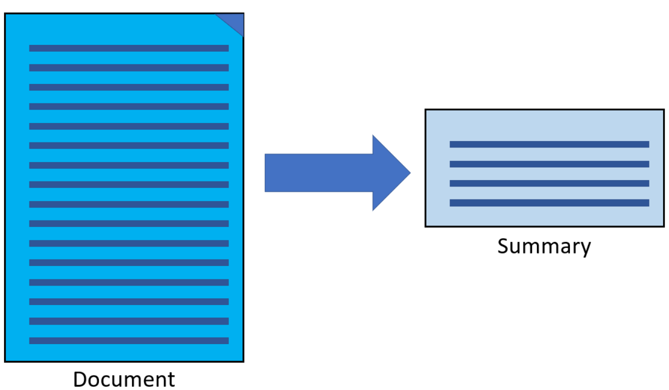

Summarization produces a summary of text by extracting sentences that collectively represent the most important or relevant information within the original content.

This feature condenses articles, papers, or documents down to key sentences.

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Language –*Text summarization*

Sentence 1

We're delighted to announce that Cognitive Service for Language service now supports extractive summarization!

Rank Score

100.00%

Sentence 2

Document summarization is a feature that produces a text summary by extracting sentences that collectively represent the most important or relevant information within the original content.

Rank Score

39.00%

Sentence 3

Extractive summarization condenses articles, papers, or documents to key sentences.

Rank Score

96.00%

Original text

We're delighted to announce that Cognitive Service for Language service now supports extractive
Rank Score: 100.00%
summarization! In general, there are two approaches for automatic document summarization: extractive and abstractive. This feature provides extractive summarization. Document summarization is a
Rank Score: 39.00%
feature that produces a text summary by extracting sentences that collectively represent the most important or relevant information within the original content. This feature is designed to shorten content that could be considered too long to read. Extractive summarization condenses articles,
Rank Score: 96.00%
papers, or documents to key sentences.

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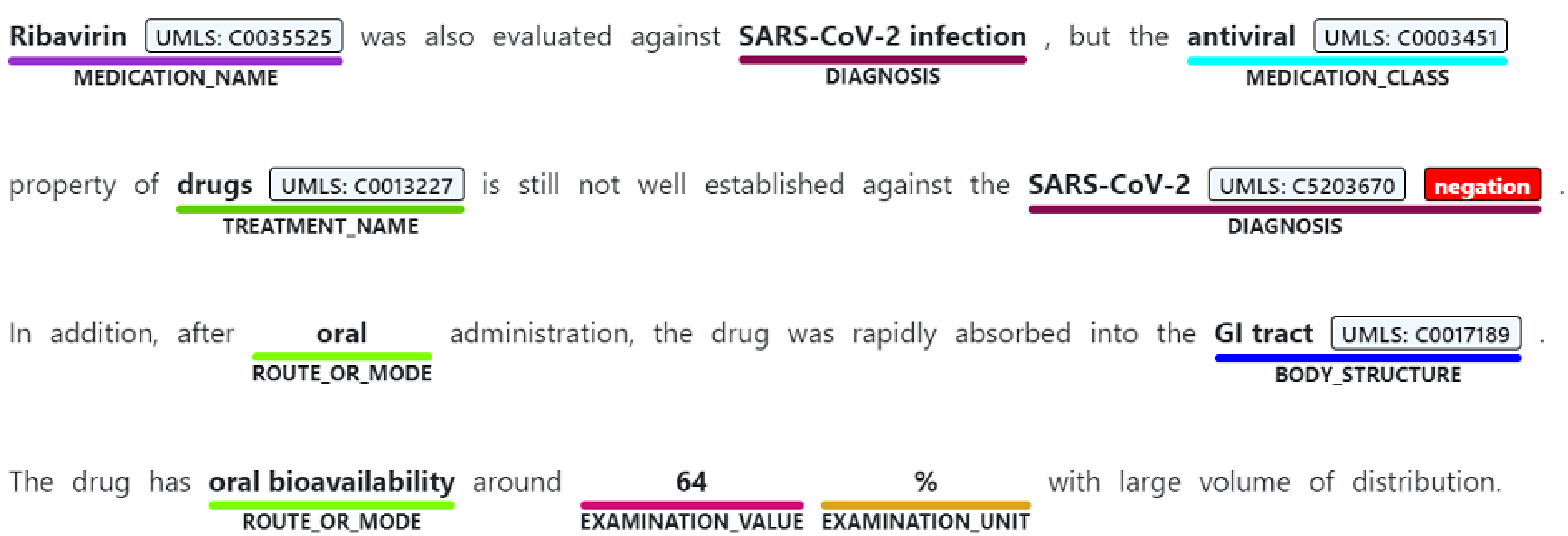
Language –*Text analytics for health*

- **Text analytics for health** is a preconfigured feature that extracts and labels relevant medical information from unstructured texts such as doctor's notes, discharge summaries, clinical documents, and electronic health records.
- **Text Analytics for health** can receive unstructured text in English, German, French, Italian, Spanish, Portuguese, and Hebrew.



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Language –*Text analytics for health*

Usage scenarios:

- **Assisting and automating** the processing of medical documents by proper medical coding to ensure accurate care and billing.
- **Increasing the efficiency** of analyzing healthcare data to help drive the success of value-based care models similar to Medicare.
- **Minimizing healthcare provider effort** by automating the aggregation of key patient data for trend and pattern monitoring.
- **Facilitating and supporting the adoption of HL7** standards for improved exchange, integration, sharing, retrieval, and delivery of electronic health information in all healthcare services.

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Language – *Text analytics for health*

Example use cases:

Use case	Description
Extract insights and statistics	Identify medical entities such as symptoms, medications, diagnosis from clinical and research documents in order to extract insights and statistics for different patient cohorts.
Develop predictive models using historic data	Power solutions for planning, decision support, risk analysis and more, based on prediction models created from historic data.
Annotate and curate medical information	Support solutions for clinical data annotation and curation such as automating clinical coding and digitizing manually created data.
Review and report medical information	Support solutions for reporting and flagging possible errors in medical information resulting from reviewal processes such as quality assurance.
Assist with decision support	Enable solutions that provide humans with assistive information relating to patients' medical information for faster and more reliable decisions.

Speech

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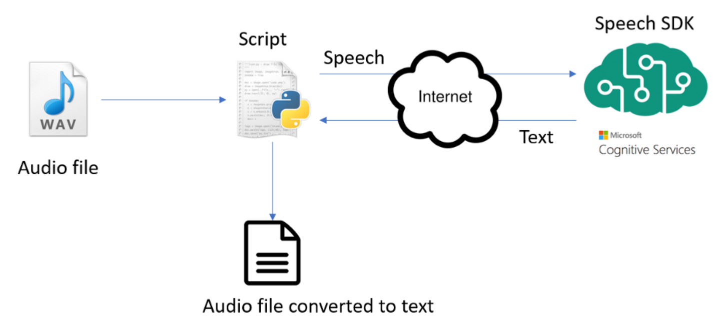
Speech –*Speech to Text*

Quickly and accurately transcribe audio to text in more than 85 languages and variants.

Customize models to enhance accuracy for domain-specific terminology.

The speech-to-text service defaults to using the Universal language model. This model was trained using Microsoft-owned data and is deployed in the cloud. It's optimal for conversational and dictation scenarios

Azure AI – Speech to Text Demo



Demo Source Code:
https://github.com/caiomsouza/Microsoft-Cognitive-Services/tree/master/speech-to-text

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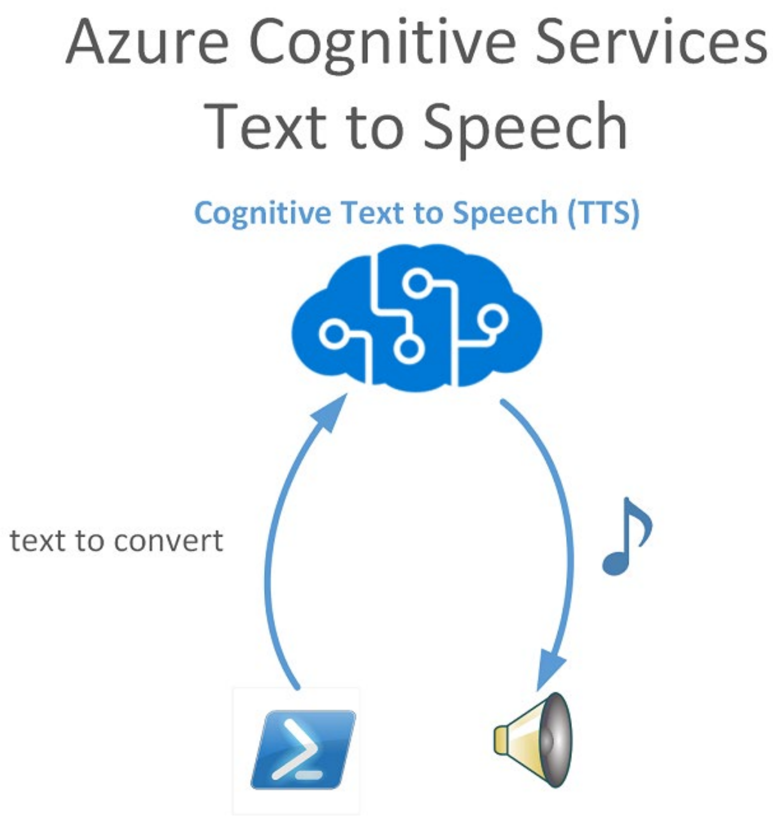
Speech –*Text to Speech*

Build apps and services that speak naturally, choosing from more than 250 voices and over 70 languages and variants.

Differentiate your brand with a customized voice, and access voices with different speaking styles and emotional tones to fit your use case—from text readers to customer support chatbots.



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Speech –*Speech Translation*

Translate audio from more than 30 languages and customize your translations for your organization’s specific terms—all in your preferred programming language.

Add high-quality translations to your apps

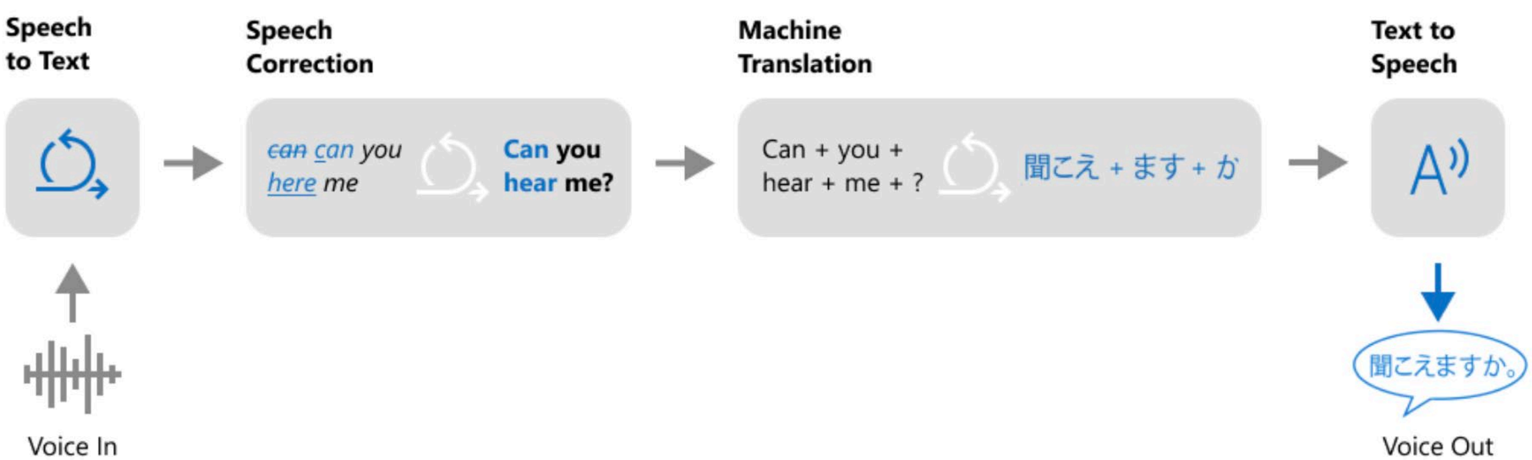
Generate speech-to-speech and speech-to-text translations with a single API call. Speech Translation captures the context of full sentences to provide accurate, fluent translations and improve communication between speakers of different languages.

Tailor translations to reflect domain-specific terminology

Customize speech recognition and translation for terminology specific to your business or industry. Train and deploy a custom translation system—without requiring machine learning expertise.

Normalize text for better translations

Speech Translation can remove verbal fillers ("um," "uh," and coughs) and repeated words, add proper punctuation and capitalization, and exclude profanities for more readable translations.



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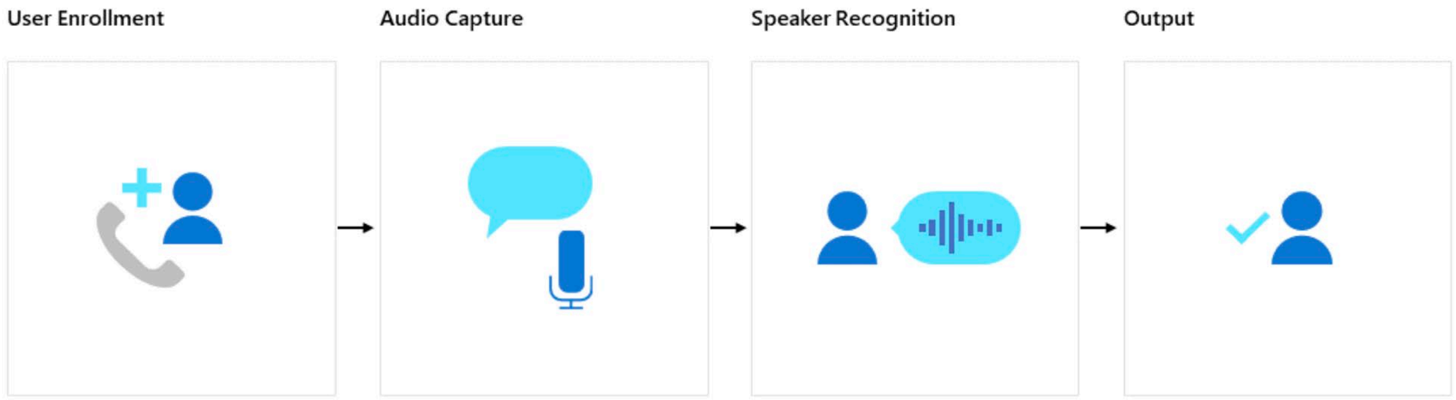
Speech –*Speaker Recognition*

Accurately verify and identify speakers by their unique voice characteristics.

The Speaker Recognition service provides algorithms that verify and identify speakers by their unique voice characteristics using voice biometry. Speaker Recognition is used to answer the question “who is speaking?”. You provide audio training data for a single speaker, which creates an enrolment profile based on the unique characteristics of the speaker's voice. You can then cross-check audio voice samples against this profile to verify that the speaker is the same person (speaker verification), or cross-check audio voice samples against a group of enrolled speaker profiles, to see if it matches any profile in the group (speaker identification).



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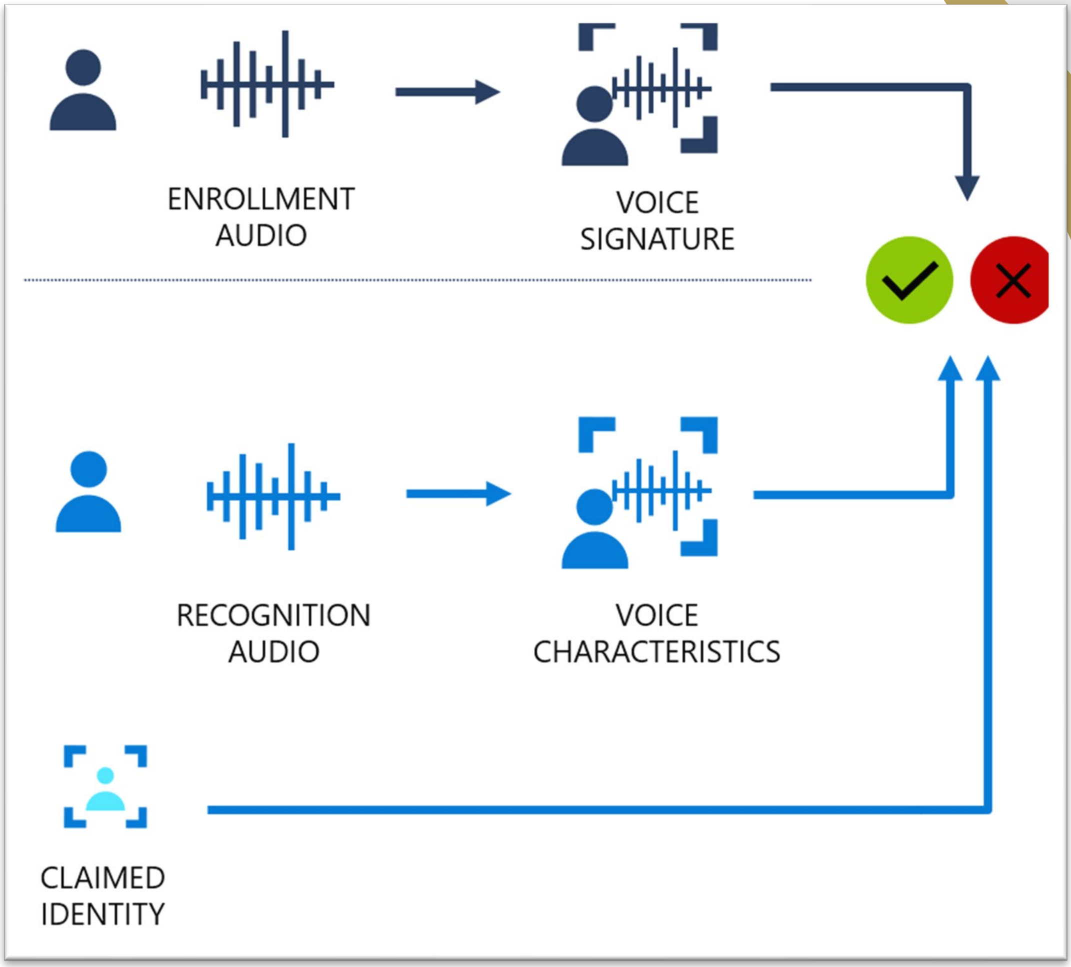
Speech –*Speaker Recognition*

Speaker Verification streamlines the process of verifying an enrolled speaker identity with either passphrases or free-form voice input.

It can be used to verify individuals for secure, frictionless customer engagements in a wide range of solutions, from customer identity verification in call centers to contact-less facility access.



Vision

Azure Cognitive Services

Azure Cognitive Services

Vision –*Computer Vision*

The cloud-based Computer Vision API provides developers with access to advanced algorithms for processing images and returning information.

By uploading an image or specifying an image URL, Microsoft Computer Vision algorithms can analyse visual content in different ways based on inputs and user choices.

Text extraction (OCR)

Extract printed and handwritten text from images and documents with mixed languages and writing styles.

Image understanding

Pull from a rich ontology of more than 10,000 concepts and objects to generate value from your visual assets.

Spatial analysis

Analyze how people move in a space in real time for occupancy count, social distancing and face mask detection.

Flexible deployment

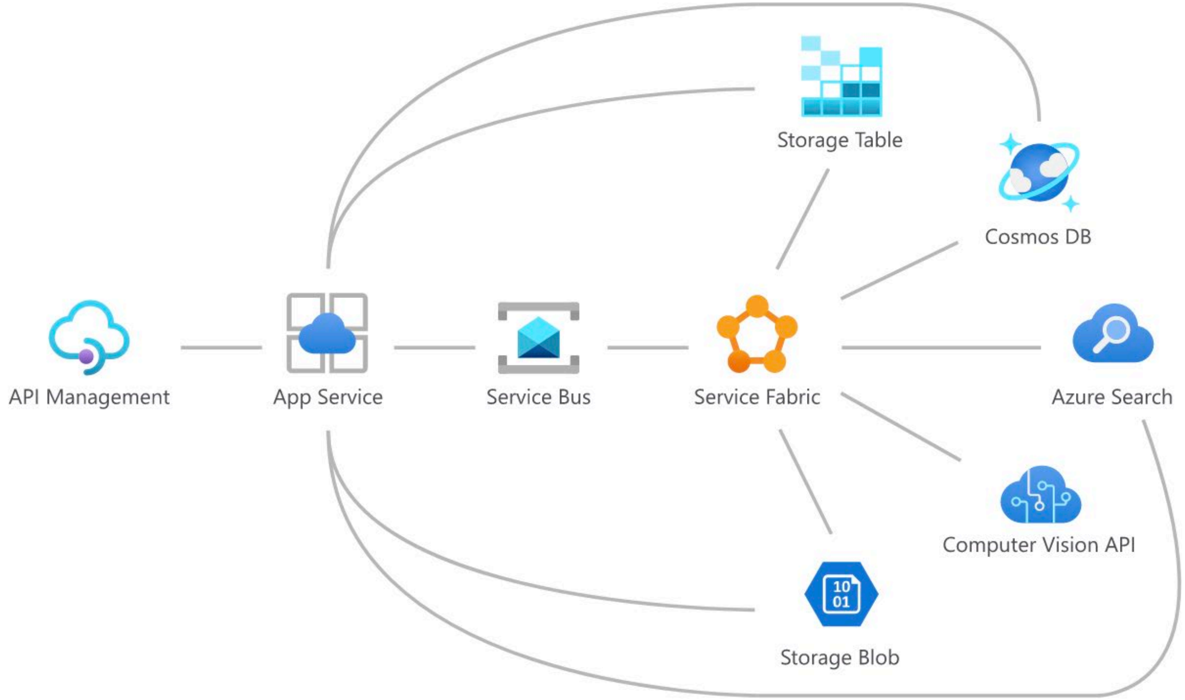
Run Computer Vision in the cloud or on the edge, in containers.

Azure Cognitive Services

Vision –*Computer Vision*

Transform your processes

Apply these Computer Vision features to streamline processes, such as robotic process automation and digital asset management.



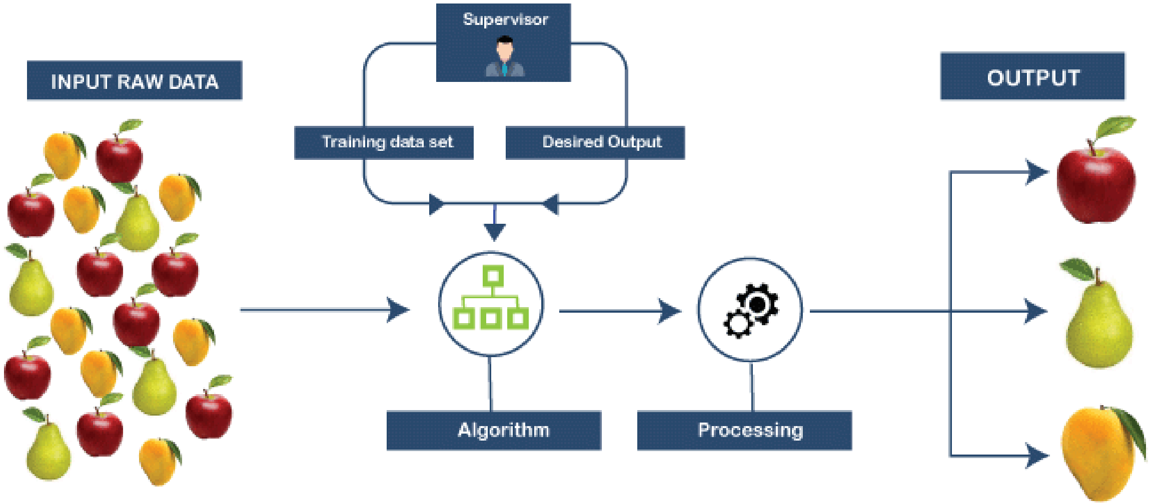
Azure Cognitive Services

Vision –*Custom Vision*

Azure Custom Vision is an **image recognition service that lets you build, deploy, and improve your own image identifier models**.

An image identifier applies labels (which represent classifications or objects) to images, according to their detected visual characteristics.

Unlike the Computer Vision service, **Custom Vision allows you to specify your own labels and train custom models to detect them**.



Azure Cognitive Services

Vision –*Face API*

Embed facial recognition into your apps for a seamless and highly secured user experience. Detect one or more human faces along with attributes such as pose, face coverings, and face location, including 27 landmarks for each face in the image.

Detect the location of a face in an image

Locate, crop, or blur faces to protect privacy using the number of faces and face rectangle coordinates. Try it on the API console.

Detect 27 landmarks for each face, including eye position

Observe potential intent using precise labels, coordinates, and 3D head pose. Learn more about face detection and attribute data.

Analyze faces for attributes, including mask detection

Determine whether someone is wearing a face mask, and whether the mask covers both the nose and mouth, with the face mask attribute. Try it on the API console.

Summary

- Content Safety
  - Prompt Shields
  - Text Analyser
  - Image Analyser

- Language
  - Named Entity Recognition (NER)
  - Custom Question Answering (Q&A Maker)
  - Text Analytics
  - Text analytics for health

- Speech
  - Speech to Text
  - Text to Speech
  - Speech Translation
  - Speaker Recognition

- Vision
  - Computer Vision
  - Custom Vision
  - Face API

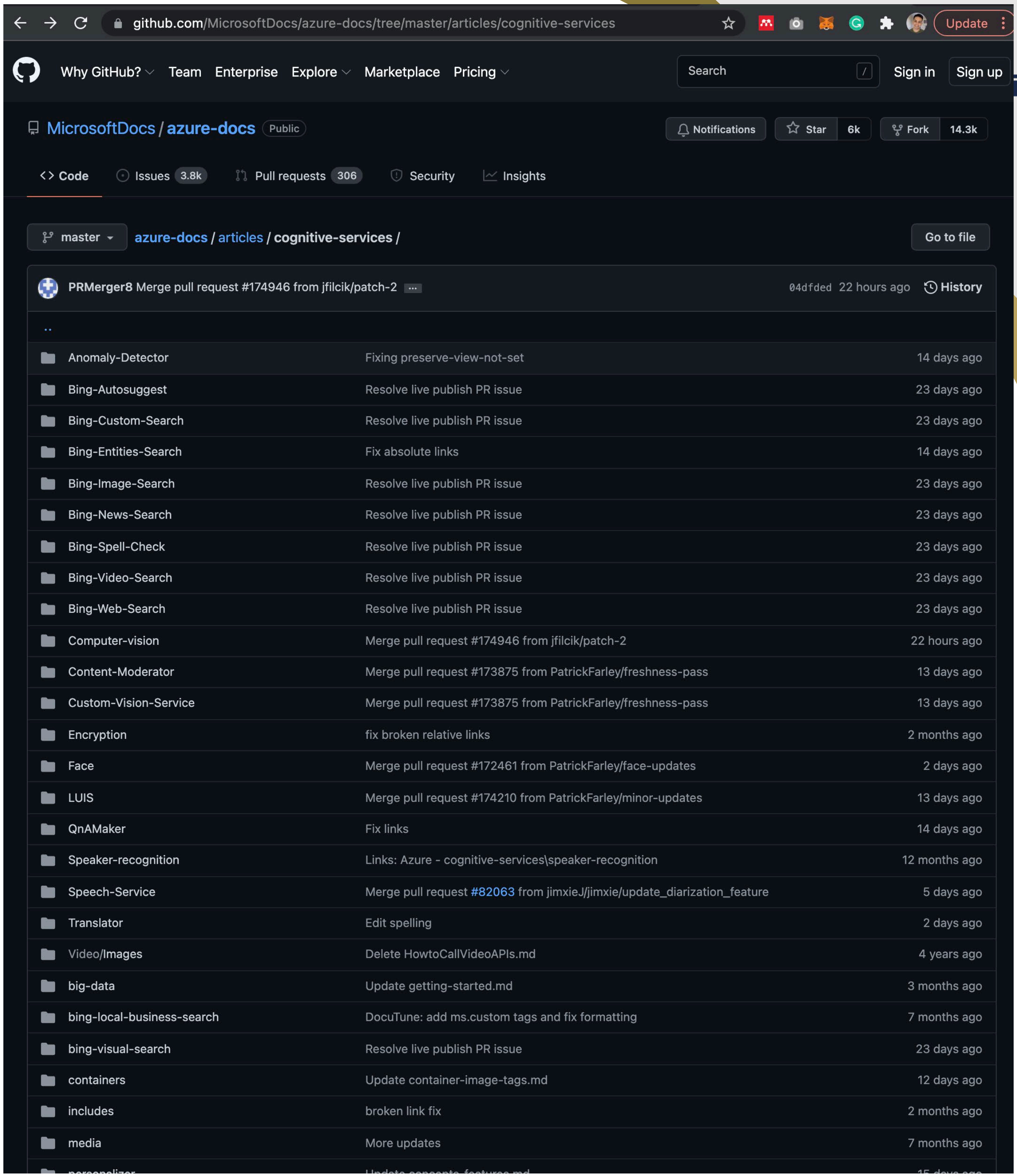
Create customizable, pretrained models built with breakthrough AI research

Deploy Cognitive Services anywhere from the cloud to the edge with containers

Get started quickly—no machine learning expertise required

Developed with strict ethical standards, empowering responsible use with industry-leading tools and guidelines

Resources



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Ulster University

Questions